Module 6

1. Pillars of Wi-Fi Security

The main pillars of Wi-Fi security are:

1. **Authentication** – verifies who is allowed to join the network.
2. **Confidentiality** – protects data using encryption so attackers cannot read it.
3. **Integrity** – ensures packets are not modified during transmission.
4. **Availability** – keeps the wireless network accessible and operational.
5. **Access Control** – controls what authenticated users are allowed to access.

These pillars together protect wireless communication from unauthorized access and attacks.

2. Difference Between Authentication and Encryption

Authentication	Encryption
Verifies identity	Protects data privacy
Answers "Who are you?"	Answers "Can others read the data?"
Happens before access	Happens during communication
Example: WPA2 password validation	Example: AES encryption

Authentication ensures only valid users join the Wi-Fi network.
Encryption converts data into unreadable form during transmission.

3. Differences Between WEP, WPA, WPA2, and WPA3

Security Type	Encryption	Security Level
WEP	RC4	Very weak
WPA	TKIP	Better than WEP

WPA2	AES-CCMP	Strong
WPA3	SAE + AES	Strongest

WEP

- Oldest Wi-Fi security
- Uses static keys
- Easily crackable

WPA

- Temporary improvement over WEP
- Introduced TKIP and MIC

WPA2

- Introduced strong AES encryption
- Uses 4-way handshake

WPA3

- Uses SAE authentication
- Better protection against password attacks
- Improved forward secrecy

4. Why WEP is Considered Insecure

WEP uses:

- Weak RC4 encryption
- Small Initialization Vector (IV)
- Static keys

Problems:

- IV reuse occurs frequently
- Attackers can capture packets and recover keys quickly

- No strong integrity protection

Because of these weaknesses, WEP can often be cracked within minutes.

5. Why WPA2 Was Introduced

WPA2 was introduced to replace weak WEP and temporary WPA security.

Main goals:

- Strong encryption
- Better integrity protection
- Secure authentication
- Enterprise-level security

WPA2 introduced:

- AES-CCMP
- Stronger key management
- Better protection against attacks

It became the standard enterprise Wi-Fi security protocol for many years.

6. Role of Pairwise Master Key (PMK) in 4-Way Handshake

The PMK is the main secret key shared between the client and AP before the handshake begins.

The PMK is derived from:

- WPA2 password (PSK mode)
  or
- 802.1X authentication process

During the 4-way handshake:

- The PMK is used to generate the PTK (Pairwise Transient Key)
- The PTK is used for encrypting unicast traffic

The PMK itself is never directly transmitted over the air.

7. How the 4-Way Handshake Ensures Mutual Authentication

The 4-way handshake proves that both:

- Client
- Access Point

know the same PMK.

Process:

1. AP sends ANonce
2. Client generates PTK using PMK
3. Client sends MIC to AP
4. AP verifies MIC
5. AP sends GTK and confirmation
6. Client verifies AP response

If both can correctly generate and verify MIC values, they trust each other.

This provides mutual authentication without sending the PMK itself.

8. What Happens if Wrong Passphrase is Entered?

If the client enters the wrong Wi-Fi password:

1. Client generates incorrect PMK

2. Incorrect PTK is generated during handshake
3. MIC validation fails
4. AP rejects the client
5. 4-way handshake fails

As a result:

- Client cannot connect
- Authentication timeout may occur
- User may see "Incorrect password"

9. What Problem Does 802.1X Solve?

802.1X solves the problem of weak shared-password security.

Without 802.1X:

- All users share the same Wi-Fi password
- Difficult to track users
- Hard to revoke individual access

802.1X provides:

- Per-user authentication
- Centralized access control
- Dynamic key generation
- Better enterprise security

It is commonly used with RADIUS servers.

10. How 802.1X Enhances Wireless Security

802.1X improves security using:

- Individual user authentication
- Dynamic encryption keys
- Centralized authentication servers

Main components:

- Supplicant (client)
- Authenticator (AP)
- Authentication Server (RADIUS)

Benefits:

- Strong enterprise authentication
- Better auditing and user tracking
- Easier access revocation
- More secure than shared PSK passwords

It is widely used in enterprise WPA2-Enterprise and WPA3-Enterprise networks.